Metric-Backed Prediction Markets
A Reputation-Capital Framework for Epistemic Coordination without Monetary Assets

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Abstract

Traditional prediction markets rely upon monetary or cryptocurrency-based capital allocation to aggregate information regarding uncertain future events. This paper introduces the Metric-Backed Prediction Market (MBPM), a prediction market in which the fundamental scarce resource is predictive performance itself rather than financial wealth.

The framework replaces monetary capital with multidimensional metric capital composed of forecasting accuracy, calibration, information gain, trust, and epistemic contribution. The resulting system forms a forecasting meritocracy whose objective function is collective knowledge production rather than financial profit maximization.

The proposed framework synthesizes concepts from economics, finance, political science, geopolitics, warfare economics, machine learning, mechanism design, information theory, and computational social choice.

The paper ends with "The End"

1 Introduction

Prediction markets have historically operated through financial incentives.

Examples include:

- Commodity futures markets.
- Election prediction markets.
- Corporate forecasting markets.
- Cryptocurrency-based forecasting protocols.

The underlying assumption is that financial capital serves as a proxy for informational confidence.

However, wealth and predictive skill are not identical quantities.

This paper proposes a market architecture in which:

"Predictive credibility becomes the settlement asset."

Instead of risking money, participants risk reputation-derived metric capital.

2 Motivation

The model attempts to solve several structural problems.

2.1 Economics

Financial markets often allocate influence according to wealth rather than informational quality.

2.2 Political Science

Public policy forecasting frequently suffers from partisan distortion.

2.3 Geopolitics

Strategic forecasting concerning wars, alliances, sanctions, and state behavior is often performed by actors possessing unequal resources.

2.4 Artificial Intelligence

Large-scale forecasting systems increasingly resemble collective intelligence systems rather than financial exchanges.

3 Metric Capital

Definition 1. *For participant i, define the metric state vector*

$$M_i = \begin{bmatrix} A_i & C_i & R_i & I_i & T_i \end{bmatrix}$$

where

A_i *Accuracy.*

C_i *Calibration.*

R_i *Resolution.*

I_i *Information gain.*

T_i *Trust score.*

The participant's total metric capital is

$$K_i = w_A A_i + w_C C_i + w_R R_i + w_I I_i + w_T T_i$$

subject to

$$\sum_j w_j = 1.$$

4 Metric Wealth Function

The market's wealth distribution becomes

$$\mathcal{K} = \{K_1, K_2, \ldots, K_n\}.$$

Unlike conventional economies,

$$K_i \neq \text{Money}.$$

Rather,

$$K_i = \text{Forecasting Merit}.$$

5 Prediction Market Mechanics

Consider event e.

Participant i reports probability

$$p_{i,e} \in [0, 1].$$

The realized outcome is

$$o_e \in 0, 1.$$

The participant commits stake

$$s_i = \alpha K_i$$

where

$$0 < \alpha < 1.$$

6 Proper Scoring Rules

The Brier score is

$$B_i = (p_i - o)^2.$$

The metric-capital update rule is

$$K_i^{t+1} = K_i^t + s_i(1 - B_i) - s_i B_i.$$

Simplifying,

$$K_i^{t+1} = K_i^t + s_i(1 - 2B_i).$$

7 Information-Theoretic Foundation

Define the market prior

$$P_0.$$

Participant forecast

$$p_i.$$

Observed outcome

$$o.$$

The information contribution is

$$IG_i = \log\left(\frac{L(o|p_i)}{L(o|P_0)}\right).$$

Positive values indicate epistemic improvement.

8 Market Consensus

The aggregate market probability is

$$P_M = \frac{\sum_i K_i p_i}{\sum_i K_i}.$$

Thus influence is proportional to demonstrated predictive performance.

9 Theorem on Convergence

Theorem 1. *Suppose:*

1. *Forecasts are scored by a strictly proper scoring rule.*
2. *Metric capital updates are monotonic in expected forecasting quality.*
3. *Event realizations are unbiased observations.*

Then the weighted consensus forecast converges toward the forecast generated by the highest-skill forecasters.

Proof. Under a strictly proper scoring rule, truthful probability reporting maximizes expected reward.
Let

$$q_i$$

denote forecasting skill.
Expected capital growth satisfies

$$E[\Delta K_i] = f(q_i)$$

where

$$f'(q_i) > 0.$$

Over repeated observations,

$$K_i \propto \exp(f(q_i)t).$$

Therefore

$$W_i = \frac{K_i}{\sum_j K_j}$$

assigns asymptotically larger weight to superior forecasters.
Consequently,

$$P_M = \sum_i W_i p_i$$

converges toward forecasts generated by the highest-quality predictors.

$\square$

10 Political Economy Interpretation

The MBPM can be viewed as a new political economy.

$$\text{Capitalism} \rightarrow \text{Financial Capital}$$

$$\text{Technocracy} \rightarrow \text{Expertise Capital}$$

$$\text{MBPM} \rightarrow \text{Predictive Capital}$$

Influence becomes contingent upon empirical forecasting performance.

11 Geopolitical Applications

Potential applications include:

- Interstate conflict forecasting.
- Alliance stability estimation.
- Nuclear deterrence analysis.
- Sanctions effectiveness forecasting.
- Energy security projections.
- Climate-security interaction forecasting.

12 Warfare Economics

Modern warfare increasingly depends upon uncertainty.
Examples include:

- Mobilization timing.
- Attrition rates.
- Industrial production.
- Supply-chain resilience.
- Technological diffusion.

The MBPM can allocate strategic credibility according to forecasting success.
Define warfare forecasting utility

$$U_W = \beta_1 A + \beta_2 I + \beta_3 R.$$

Strategic planners maximize

$$\max U_W.$$

13 Artificial Intelligence Layer

An AI forecasting agent is represented as

$$\pi_\theta(x) = P(Y|X = x; \theta).$$

Metric capital for AI system j becomes

$$K_j^{AI} = \sum_t \gamma^t R_t.$$

The market thereby forms a reinforcement-learning environment.

14 Machine Learning Interpretation

The market acts as a distributed ensemble.
Consensus prediction:

$$P_M = \sum_i W_i p_i.$$

This is analogous to:

- Boosting.
- Bayesian model averaging.
- Mixture-of-experts architectures.

15 Algorithm

Pseudo-Code

```
Initialize K_i

For each event e:

 Receive forecasts p_i

 Compute market probability P_M

 Observe outcome o

 Compute Brier score

 Update K_i

 Recompute weights W_i

Repeat
```

16 Computational Complexity

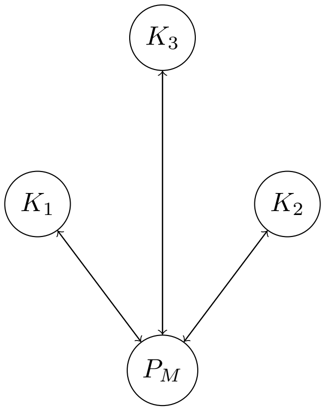
For n participants and m events,

$$T(n, m) = O(nm).$$

Storage requirement:

$$S(n, m) = O(n + m).$$

17 Network Representation



The graph represents recursive credibility accumulation.

18 Metric Asset Classes

Asset	Interpretation	Function
Accuracy	Correctness	Reward precision
Calibration	Reliability	Penalize overconfidence
Resolution	Sharpness	Reward discrimination
Information Gain	Novel insight	Reward discovery
Trust	Governance	Reward integrity

Table 1: Metric Capital Components

19 Meta-Market Extension

Metrics themselves become tradable informational assets.

Define metric vector

$$m_i = (m_{i1}, m_{i2}, \ldots, m_{ik}).$$

Market valuation:

$$K_i = \sum_j q_j m_{ij}.$$

The market evolves into a market for forecasting skill itself.

20 Philosophical Implications

The framework operationalizes a novel epistemic principle:

> Influence should emerge from demonstrated predictive success rather than inherited wealth, political authority, or institutional prestige.

The MBPM therefore constitutes a formal mechanism for epistemic meritocracy.

21 Conclusion

The Metric-Backed Prediction Market replaces monetary settlement with reputation-based metric capital.

The resulting system unifies:

- Economics,
- Finance,
- Information theory,
- Artificial intelligence,
- Political science,
- Geopolitics,
- Warfare economics,
- Computational social choice.

The central innovation is the transformation of predictive performance into the primary economic asset.

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Glossary

Metric Capital
Forecasting wealth derived from measurable performance.

Prediction Share
Influence unit derived from metric capital.

Information Gain
Epistemic contribution relative to consensus.

Trust Score
Integrity-weighted credibility measure.

Predictive Capitalism
An economy based on forecasting performance.

Epistemic Meritocracy
A governance structure where influence follows predictive success.

The End